

Weekly reflections

This week focused on understanding the dataset and developing a structured approach to automatically classify Reddit posts related to the food truck business. I began by exploring the dataset in detail to understand the type of content available, the structure of the text fields, and the existing annotations. This initial exploration helped clarify the main objective: identifying posts from users who are seeking help before starting a food truck versus those who already operate a food truck and need assistance with operational issues.

After discussing the problem with my manager, we agreed that directly classifying posts into these two high-level categories would be too coarse. Instead, I was advised to introduce subcategories, as they provide more granular and unique signals within the text. Based on this guidance, I designed a set of startup-related subcategories (such as Licensing & Permits and Business Planning) and operational subcategories (such as Cold Storage and Equipment Maintenance).

To enable automatic classification, I created a keyword-based system where each keyword was assigned an arbitrary numeric value between 1 and 100, representing the strength of its association with a given subcategory. These values are not machine learning weights but manually assigned scores used to quantify how strongly a keyword indicates startup or operational intent. This approach was chosen to prioritize explainability and control over a black-box model.

Using this structure, I implemented a Python-based batch classification function that processes the full dataset. For each post, the system scans the text, aggregates keyword scores into startup and operational totals, and assigns a final label: Pre-Startup Help, Operational Help, or Unclear/Mixed when the intent is ambiguous. This logic was applied across the entire dataset automatically.

Overall, this week's work resulted in a transparent, rule-based classification system that transforms unstructured Reddit posts into structured intent data. The approach balances technical rigor with interpretability and provides a strong foundation for future enhancements, such as expanding keyword lists or integrating machine learning models.

Planned Work for the Coming Week

In the coming week, I plan to refine and expand the current classification system. This will include extending the keyword lists for each subcategory to improve coverage and reduce the number of posts labeled as mixed or unclear. I also plan to experiment with adjusting scoring thresholds and incorporating keyword frequency to improve classification sensitivity.

Additionally, I aim to evaluate the rule-based approach by comparing its outputs against existing labels in the dataset and identifying areas of mismatch. Based on these findings, I will explore

the possibility of integrating lightweight machine learning techniques, such as TF-IDF-based models, to complement the rule-based system. Finally, I plan to generate summary statistics and visualizations to better understand the distribution of startup versus operational intent across the dataset.